**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 059 · Tracing Programming Constructs

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. When is a condition-controlled loop suitable?

Q2. Linear-search [7, 2, 9, 4] for 4. How many comparisons are needed?
A. 1
B. 2
C. 4
D. 3

Q3. “An IF statement must execute every possible branch regardless of its condition.” Correct this. Explain why.

Q4. Do one bubble-sort pass from left to right, smallest to largest, on [6, 1, 5, 3]. Show the array after each comparison. Which value ends up on the right?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Trace table — Records the current variable values after relevant instructions.

Selection tracing — Only the true branch of an IF/ELSE is followed.

Iteration tracing — Record the changes for every repeated execution.

Debugging — Compare expected and traced results to find the first incorrect step.

Algorithm — A sequence of precise problem-solving instructions.

Trace table — A table tracking variable values during execution.

## New learning

Trace table: Records the current variable values after relevant instructions. For example, consider a score calculator.

Selection tracing: Only the true branch of an IF/ELSE is followed. For example, consider a discount total.

Iteration tracing: Record the changes for every repeated execution. For example, consider a lesson attendance.

Debugging: Compare expected and traced results to find the first incorrect step. For example, consider a countdown.

## Worked example

A score calculator is being developed. Explain how trace table and selection tracing are used. [4 marks]

1. Identify where this particular program needs control or values.

2. Trace table [1]; Records the current variable values after relevant instructions. [1].

3. Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1].

4. Give one accurate point and a linked reason for each technique.

# We Do · Practise with guidance

### Guided 1

A discount total is being programmed. Explain why trace table matters. [2 marks]

Worked method: Trace table [1]; Records the current variable values after relevant instructions. [1].

### Guided 2

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

For a lesson attendance, explain how selection tracing is used. [2 marks]

Worked method: Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1].

**Guided 3**

A countdown needs iteration tracing and debugging. Explain the roles of both. [4 marks]

Worked method: Iteration tracing [1]; Record the changes for every repeated execution. [1]. Debugging [1]; Compare expected and traced results to find the first incorrect step. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

Predict the value printed after the loop.

```
totalGB = 0
for i = 1 to 3
    totalGB = totalGB + i
next i
print(totalGB)
```

predict

### Question 1

State TWO concepts needed for this programming technique. [2 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

### Question 2

Explain how selection tracing and iteration tracing can help a points accumulator. [4 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

### Question 3

Explain the roles of trace table, iteration tracing and debugging when creating a fitness tracker. [6 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Exit Ticket · Answer before checking

A trace table records the values of variables after relevant instructions execute.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. When repetition continues or ends according to a condition, such as valid input.

Q2. C — 4

Q3. It chooses a branch using a condition. Selection can be nested.

Q4. (a) [[1, 6, 5, 3], [1, 5, 6, 3], [1, 5, 3, 6]]; largest value 6 is at the right end.

## I Do / We Do pages · Guided practice

Q1. Trace table [1]; Records the current variable values after relevant instructions. [1].

Q2. Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1].

Q3. Iteration tracing [1]; Record the changes for every repeated execution. [1]. Debugging [1]; Compare expected and traced results to find the first incorrect step. [1].

## You Do pages · Code First

```
totalGB = 0
for i = 1 to 3
    totalGB = totalGB + i
next i
print(totalGB)
```

Expected result: 6

## You Do pages · Independent questions

Q1. Trace table [1]; Iteration tracing [1].

Q2. Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1]. Iteration tracing [1]; Record the changes for every repeated execution. [1].

Q3. Trace table [1]; Records the current variable values after relevant instructions. [1]. Iteration tracing [1]; Record the changes for every repeated execution. [1]. Debugging [1]; Compare expected and traced results to find the first incorrect step. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**